

KIM HYEONGHAN

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Unified Astronomy Thesaurus (UAT) keywords:

Large-scale structure, gravitational lensing, dark matter distribution, intracluster medium

EDUCATION

PhD of Science, Astronomy <i>Major: Astronomy</i> Yonsei University; Advisor: Prof. M. James Jee	Aug 2025 Seoul, KR
Bachelor of Science, Astronomy <i>Major: Astronomy</i> Yonsei University	Feb 2018 Seoul, KR

EMPLOYMENT

Post-doctoral researcher Duke University; PI: Prof. M. Troxel	Sep 2025 – NC, USA
Korea Astronomy and Space science Institute (KASI) Student Researcher Supervisor: Dr. B. W. Sohn and Dr. M. Martínez-Paredes • Martínez-Paredes, M., González-Martín, O., HyeongHan, K. , et al. 2021, <i>ApJ</i> , 922, 157	Jan 2019 – Aug 2020 Daejeon, KR
Korean VLBI Network (KVN) System Assistant Yonsei University • KVN maintenance	Jul – Aug 2016 Seoul, KR

UNDER-REVIEW & REFEREED PUBLICATIONS

First and/or corresponding[†] author

- [6] **HyeongHan, K.**, Finner, K., Jee, M. J., et al. *Dark and Luminous Matter in the Coma Cluster: Probing Galaxy Cluster Assembly Through Filaments with Weak Lensing and Multiwavelength Observations*. submitted to *A&A*, arXiv:2606.12523
- [5] Kim, S., **HyeongHan, K.**[†], Lee, W., et al. *Weak-Lensing Analysis of the Galaxy Cluster Abell 85: Constraints on the Merger Scenarios of Its Southern Subcluster*. *ApJ*, 1003, 16 (May 2026)
- [4] **HyeongHan, K.**, Jee, M. J., W. Lee et al. *Direct Evidence of a Major Merger in the Perseus Cluster*. *Nature Astronomy*, 9, 925 (Jun 2025)
- [3] **HyeongHan, K.**, Jee, M. J., Cha, S. et al. *Weak-Lensing Detection of Intracluster Filaments in the Coma Cluster*. *Nature Astronomy*, 8, 377 (Mar 2024)
- [2] **HyeongHan, K.**, Cho, H., Jee, M. J. et al. *Weak-Lensing Analysis of the Complex Cluster Merger Abell 746 with Subaru/Hyper Suprime-Cam*. *ApJ*, 962, 100 (Feb 2024)
- [1] **HyeongHan, K.**, Jee, M. J., Rudnick, L. et al. *Discovery of a Radio Relic in the Massive Merging Cluster SPT-CL J2023-5535 from the ASKAP-EMU Pilot Survey*. *ApJ*, 900, 127 (Sep 2020)

Second/third author

- [5] Pal, A., Kale, R., **HyeongHan, K.**, et al. *A Multiscale Spectro-polarimetric View of the Radio Relics in the Merging Galaxy Cluster MACS J1752+4440 with uGMRT and EVLA*. submitted to *ApJ*
- [4] Jiménez-Teja, Y., Román, J., **HyeongHan, K.**, et al. *Deep view of the intracluster light in the Coma cluster of galaxies*. *A&A*, 694, 216 (Feb 2025)
- [3] Cha, S., **HyeongHan, K.**, Scofield, Z. P. et al. *Precision MARS Mass Reconstruction of Abell 2744: Synergizing the Largest Strong Lensing and Densest Weak Lensing Datasets from JWST*. *ApJ*, 961, 186 (Feb 2024)
- [2] Martínez-Paredes, M., González-Martín, O., **HyeongHan, K.** et al. *Modeling the Unresolved NIR-MIR SEDs of Local ($z < 0.1$) QSOs*. *ApJ*, 922, 157 (Dec 2021); Contribution: IR photometry and modeling

- [1] Finner, K., **HyeongHan, K.**, Jee, M. J. et al. *Exemplary Merging Clusters: Weak-lensing and X-Ray Analysis of the Double Radio Relic, Merging Galaxy Clusters MACS J1752.0+4440 and ZWCL 1856.8+6616*. ApJ, 918, 72 (Sep 2021)

Contributing author

- [13] Finner, K. et al. (incl. **HyeongHan, K.**) *Near-IR Weak-lensing (NIRWL) Measurements in the CANDELS Fields. II. Mass Mapping and Overdensity Characterization*. submitted to ApJ arXiv:2605.15264
- [12] Cho, B. Y. et al. (incl. **HyeongHan, K.**) *Joint JWST–DECam Lensing Reveals That the Bullet Cluster Is a Minor Merger*. ApJ, 1005, 28 (Jul 2026)
- [11] Wang, Q. et al. (incl. **HyeongHan, K.**) *Reproducing Abell 2744 with the HyperMillennium Simulation*. MNRAS, 548, 540 (May 2026)
- [10] Jain, S. et al. (incl. **HyeongHan, K.**) *SDSS-C4 3028: the Nearest Blue Galaxy Cluster Devoid of an Intracluster Medium*. OJAp, DOI: 10.33232/001c.144911 (Sep 2025)
- [9] Cha, S. et al. (incl. **HyeongHan, K.**) *A High-Caliber View of the Bullet Cluster Through JWST Strong and Weak Lensing Analyses*. ApJL, 987, L15 (June 2025)
- [8] Finner, K. et al. (incl. **HyeongHan, K.**) *Weak-Lensing Characterization of the Dark Matter in 29 Merging Clusters that Exhibit Radio Relics*. ApJS, 277, 28 (Mar 2025)
- [7] Rajpurohit, K. et al. (incl. **HyeongHan, K.**) *Abell 746: A highly disturbed cluster undergoing multiple mergers*. ApJ, 966, 38 (May 2024)
- [6] Lee, W. et al. (incl. **HyeongHan, K.**) *Discovery of a large-scale bent radio jet in the merging cluster Abell 514*. ApJL, 957, 4 (Nov 2023)
- [5] Finner, K. et al. (incl. **HyeongHan, K.**) *Near-IR Weak-Lensing (NIRWL) Measurements in the CANDELS Fields I: Point-Spread Function Modeling and Systematics*. ApJ, 958, 33 (Nov 2023)
- [4] Lee, W. et al. (incl. **HyeongHan, K.**) *Discovery of a Double Radio Relic in ZwCl1447.2+2619: A Rare Testbed for Shock-acceleration Models with a Peculiar Surface-brightness Ratio*. ApJ, 924, 18 (Jan 2022)
- [3] Kim, J. et al. (incl. **HyeongHan, K.**) *Head-to-Toe Measurement of El Gordo: Improved Analysis of the Galaxy Cluster ACT-CL J0102-4915 with New Wide-field Hubble Space Telescope Imaging Data*. ApJ, 923, 101 (Dec 2021)
- [2] Norris, R. et al. (incl. **HyeongHan, K.**) *The Evolutionary Map of the Universe pilot survey*. PASA, 38, 46 (Sep 2021)
- [1] Di Mascolo, L. et al. (incl. **HyeongHan, K.**) *Multiwavelength view of SPT-CL J2106-5844. The radio galaxies and the thermal and relativistic plasmas in a massive galaxy cluster merger at $z \simeq 1.13$* . A&A, 650, 153 (Jun 2021)

SELECTED TALKS AND POSTERS

Contributed Talks

ASKAP-EMU Virtual Meeting	Jul 2020
EMU-CCW Virtual Meeting	Sep 2020
102nd Korean Astronomical Society Meeting	Sep 2020
103rd Korean Astronomical Society Meeting	Apr 2021
240th American Astronomical Society Meeting, Pasadena, US	Jun 2022
International Astronomical Union General Assembly 2022, Busan, South Korea	Aug 2022
243rd American Astronomical Society Meeting, New Orleans, US	Jan 2023
108th Korean Astronomical Society Meeting	Oct 2023
1st Merging Cluster Workshop at Yonsei University	Dec 2023

109th Korean Astronomical Society Meeting	Apr 2024
2024 European Astronomical Society Meeting, Padova, Italy	Jul 2024
111th Korean Astronomical Society Meeting	Apr 2025
Filaments in the Multi-scale Universe at Kavli-IPMU	Oct 2026

Invited Talks

2nd Merging Cluster Workshop at Yonsei University	Nov 2025
Galaxy Cluster Meeting at CfA	Oct 2026

Posters

100th Korean Astronomical Society Meeting	Oct 2019
9th Korean Institute for Advanced Study (KIAS) Cosmology Workshop	Nov 2020
Galaxy Cluster Formation Meeting	Jun 2021
European Astronomical Society Meeting	2024
Galaxy Clusters & Radio Relics II meeting at CfA	Sep 2024
11th KIAS Cosmology Workshop	Oct 2024

HONORS AND AWARDS

Brain Korea 21 Plus Scholarship (KRW 17.5 million)	2018–2021
Brain Korea 21 Four Project Scholarship (KRW 33.6 million)	2021–2023
Yonsei Merit Academic Paper Award (KRW 1 million)	2020
Yonsei Merit Academic Student Award	2021
BLEU Fellowship (KRW 4 million)	2023
Yonsei Merit Academic Paper Award	2024

TEACHING EXPERIENCE

Mentor at Duke University, Manvik Kongwad, NCSSM → UNC	Sep 2025 - Jan 2026 NC, USA
MSc project supervisor, Soojin Kim, Seoul National University → UChicago	Nov 2024 - May 2026
Weak-Lensing Analysis of the Galaxy Cluster Abell 85: Constraints on the Merger Scenarios of Its Southern Subcluster (ApJ, 1003, 16)	Seoul, KR
Project mentor, Hye Gyeong Khim, Yonsei University	Jan 2023 - Seoul, KR
Project mentor, Boseong Y. Cho, Yonsei University	Jan 2022 - Dec 2025
Joint JWST–DECam Lensing Reveals That the Bullet Cluster Is a Minor Merger (ApJ, 1005, 28)	Seoul, KR
Understanding of space	2018, 2020
TA; Yonsei University	Seoul, KR

ACCEPTED OBSERVATIONAL PROPOSALS

7 PI proposals

X-ray: XMM-Newton AO20 (125 ks), AO21 (125 ks), AO23 (160 ks), AO25 (158 ks)

Radio: ATCA 2020OCT (23.5 hr)

Optical: Subaru/HSC 2024B (5 hr), 2025B (3 hr)

More than 17 Co-I proposals

X-ray: Chandra Cycles 22 and 24; XMM-Newton AO23–25; NuSTAR Cycle 12

Ultraviolet/optical/infrared: HST Cycle 34; JWST Cycle 3; Roman Cycle 1; Gemini-N/GMOS 2021A;

Subaru/HSC 2024A and 2026B; Subaru/PFS 2026B; MMT/Binospec and MMT/MegaCam 2022A

Radio/submillimeter: ALMA Cycle 8; uGMRT Cycles 36–38 and 41; VLA 2025A; MeerKAT 2023 and 2024

ACADEMIC SERVICE

Referee service

ApJ Letters

Department of Astronomy Social Exchange Meeting LOC (chair) May 2019

The 3rd Astro Poster Jamboree LOC Aug 2019

9th Korea-Japan Workshop on Dark Energy LOC Nov 2022

<https://sites.google.com/view/9kjwde/home?authuser=0>

Yonsei Merging Cluster Workshop 2023 LOC (chair) / SOC Dec 2023

<https://yhep-indico.yonsei.ac.kr/event/549/>

Duke Cosmology Group Journal Club organizer (chair) Jan 2026–

Collaborations

The Evolutionary Map of the Universe (EMU) conducted with Australian SKA Pathfinder (ASKAP)

The Roman Cosmology with High Latitude Imaging Survey Project Infrastructure Team (Roman HLIS PIT) for the Nancy Grace Roman Space Telescope

The Rubin Observatory Legacy Survey of Space and Time Dark Energy Science Collaboration (LSST DESC)

PUBLIC OUTREACH

YTN Science 2024

Link: <https://youtu.be/xfwcomfaOaQ?si=gQ25wE7e3H629QGN> (~6.1k views)

Sciencewithpeople 2024

Link: <https://youtu.be/49fXbZZPGx0?si=yTWZp6MbLB2gnkdF> (~19k views)

Sciencewithpeople 2025

Link: <https://youtu.be/E-HVNDdaCjs?si=K75h6Gl1no9j7qYi> (~9.7k views)

REFERENCES

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Prof. Emer. Lawrence Rudnick University of Minnesota

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